A Research Framework for Virtual-Reality Neurosurgery Based on Open-Source Tools

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Abstract

Fully immersive virtual reality (VR) has the potential to improve neurosurgical planning. However, there is a lack of research tools for this area. We present a research framework for VR neurosurgery based on open-source tools. We showcase the potential of such a framework using clinical data of two patients and research data of one subject. As first step toward practical evaluations, certified neurosurgeons positively assessed the VR visualizations and interactions using head-mounted displays. Methods and findings described in our study thus provide a foundation for research and development of versatile and user-friendly VR tools for improving neurosurgical planning and training.

Keywords: Virtual reality, Imaging, Health informatics, Image segmentation, Medical technologies, 3D imaging.

Index Terms: Human-centered computing—Interaction paradigms—Virtual reality; Human-centered computing—Interaction devices—Haptic devices;

1 Introduction

As of 2019, neurosurgeons still plan many interventions using 2D images on 2D screens to build an internal 3D representation of what to expect during surgery and of how to proceed to achieve best possible outcomes. However, this process is time-consuming and requires extensive experience. Virtual Reality (VR) offers the possibility for an immersed experience of 3D content, and thus gives neurosurgeons the means of a direct 3D perception of and interaction with the anatomy relevant for planning. So far, while first commercial products are currently being developed, an open-source immersive VR framework is not available for this task, nor therefore, for research and development in this application context. However, using open-source tools in research is important to ensure that the work can be reproduced properly and can be used as a foundation for further work by the community. As a consequence, we have started to develop a research framework for VR Neurosurgery based on open-source tools and report our preliminary results.

2 Related Work

Visualization Several research groups successfully implemented realistic 3D VR visualizations of brain structures [1, 7, 9, 12]. Most recently, [10, 11] exploited state of the art computer graphics and digital modeling technologies to create intricate, anatomically realistic and visually appealing virtual models of the cranial bones and cerebrovascular structures.

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Commercial frameworks Several VR integrated hardware and software platforms have been established in recent years in the field of neurosurgery. The Dextroscope® has been used in a variety of intervention simulations [13, 15, 17–19]. NeuroTouch incorporates haptic feedback designed for the acquisition and assessment of technical skills involved in craniotomy-based procedures and tumor resection [3, 4, 6]. ImmersiveTouch® provides platforms in AR and immersive VR environments with haptic feedback for neurosurgical training [2, 14]. Commercial platforms lacking scientific assessment include NeuroVR, Surgical Theater, Osso VR and Touch Surgery. A recent review regarding the development and utilization of VR environments for neurosurgical purposes can be found in [16].

3 Methods

3.1 Data & Segmentation

We have used the MRI and CT data of one healthy subject (S1, MRI only) and 2 patients (P1-2) as our pilot cohort to construct detailed 3D models to be viewed and interacted with in VR. To produce high-quality results on the clinical routine data, we adapt the segmentation pipeline used to create the model of S1 [8] for each patient individually. FSL FLIRT, FreeSurfer, 3D Slicer and our own code are used to segment the data. When running on an i7-8700K CPU (12 3.7 GHz threads), the semi-automated part of the pipeline (without manual segmentation) generally takes 6-7 h to complete, with FreeSurfer taking 5-6 h.

3.2 Visualization & User Interface

We developed our VR environment using the Unity3D game engine. We use 2 directional real-time light sources , set environment lighting ambient mode to real-time global illumination, the standard Unity3D shader, perspective projection rendering and forward rendering. An Oculus Riftheadset and two Touch controllers bundle were used for a fully immersive VR environment. User interface (UI) elements holding the layers' names and control modes are generated inside a canvas that can be switched on/off. Two modes of interaction were implemented , the raycaster interaction (RI) mode and the controllers tracking interaction (CTI) mode. In RI mode, UI elements in the canvas additionally includes control modes, namely, Move Avatar, which enables translational movement in three perpendicular axes and Rotate Model, which allows yaw and pitch rotations. The Scale Model UI element enables the user to scale the model up and down. Finally, a Reset option is provided to return the model and the avatar to their initial state. The CTI mode is based on the distance and the relative angles between the controllers, allowing yaw, pitch, roll rotations and model scaling, by creating circular and linear motions with the controllers, respectively. CTI interactions have been suggested to us by the experienced neurosurgeons having preliminarily evaluated the framework. A cross-sectional view of the model is provided by controlling three planes.

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4 RESULTS

Videos demonstrating our results are available on our website `https://www.tnt.uni-freiburg.de/research/medical-vr-segmentation/aveh_2019`. All models can be extended by adding more segmented areas or structures. Not all structures were included into the VR representation of P1-2.

4.1 Segmentations

Subject 1 Cerebrovascular tree, skull, CSF, dura mater, eyes, fat, gray matter, white matter, internal air, skin, and soft tissue were segmented and modeled.

Patient 1 Meninges, catheters, optic tract, MRA vessels, CTA arteries, skull, tumor, eyes, cerebral and cerebellar gray and white matters, subcortical structures, ventricles and brainstem were segmented and modeled.

Patient 2 Intracranial arteries, skull, tumor, cerebral and cerebellar gray and white matters, subcortical structures, ventricles and brainstem and brainstem nerves were segmented and modeled.

4.2 Preliminary Performance Evaluation

We evaluated the performance of the VR visualizations by recording the frame delta time and the count of vertices passed to the GPU for different model configurations and interactions. A NVIDIA GTX 1080 Ti is used as GPU. Frame rates never dropped below 30 FPS which ensures that the interaction stays smooth under any load.

4.3 Preliminary Physician Evaluation

Two experienced neurosurgeons inspected the 3D models postoperatively in VR prior to inspecting the 2D images in 2D. They reported being surprised by the accuracy of the models, the degree of immersion and the speed at which they were able to assimilate the information. During the subsequent inspection of the 2D images, the neurosurgeons were able to quickly verify their VR-based internal 3D representation and successfully transferred these insights from VR to 2D.

5 DISCUSSION

In the present study we, to the best of our knowledge, for the first show that existing open-source tools and consumer hardware can be used to create a research framework for VR neurosurgery. In parallel to ongoing development by commercial solutions, we believe that there is also a need for academic research into this topic. We anticipate that the possibility of using open-source tools will facilitate future research and development into VR applications in the context of neurosurgical planning, as well as teaching and training.

We have show-cased the working procedure of our framework on data of one healthy subject and two tumor patients. Due to different requirements in regard to the final models, the processing pipeline had to be personalised. For automated segmentation and modeling, this need for personalisation represents a big challenge. In order to solve this issue, we plan to apply state-of-the-art machine learning techniques, e.g., deep convolution neural networks, in the future.

To further improve our framework, we plan to devise a more intuitive, user-friendly and interactive UI, and implement mesh manipulations to mimic neurosurgical procedures, as demonstrated by [5].

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